

### Lecture 3

- In practice, sometimes matrices are not enough, we need multidimensional arrays!

e.g. : Python reads an image with  $n \times m$  pixels as an array of size  $(n, m, 3)$

- Here, we give an intensity value to each pixel regarding the colors R, G, B.

example 2: Maximum daily temperature  $T$  in the past 10 years.

- What if we have multiple pictures?

•  $10^4$  pictures of  $1920 \times 1080$  pixels

$\Rightarrow$  can be represented with a 4-dim array:

$(10^4, 1920, 1080, 3)$

when is reshape useful?

Suppose we have an RGB array of size  $(n, m, 3)$ .

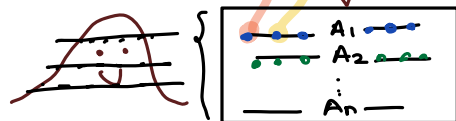
We can reshape it into a matrix s.t each row represents a pixel:

$$\begin{array}{l} \text{Pixel 1: } [R \ G \ B] \\ \text{Pixel 2: } [R \ G \ B] \\ \vdots \\ \text{Pixel } n \times m: [R \ G \ B] \end{array} \Rightarrow \text{new array has size } (n \cdot m, 3)$$

(How? use `image.reshape(n * m, 3)`)

originally:  $\begin{bmatrix} [A_1] \\ [A_2] \\ \vdots \\ [A_m] \end{bmatrix}$

$\hookrightarrow$  each matrix represents a "row" of our picture



where

$$A_i = \begin{bmatrix} R & G & B \\ R & G & B \\ \vdots & \vdots & \vdots \\ R & G & B \end{bmatrix} \text{ size } (m, 3)$$

each row of  $A_i$ , a different pixel in row  $i$  of picture.

Since reshape preserves order row-wise

↓  
new matrix after reshape:  
(n·m, 3)

|         |           |           |
|---------|-----------|-----------|
| [R G B] | 1st pixel | (1st row) |
| [R G B] | 2nd pixel | (1st row) |
| ⋮       |           |           |
| [R G B] | mth pixel | (1st row) |
| [R G B] | 1st pixel | (2nd row) |
| [R G B] | 2nd pixel | (2nd row) |
| ⋮       |           |           |

and so on.

### Arrays in Python: multiplication

Matrix-matrix multiplication using matmul:

• If  $A \in \mathbb{R}^{d_1 \times d_2 \times d_3}$  and  $B \in \mathbb{R}^{d_1 \times d_3 \times d_4}$   
 $\Rightarrow AB \in \mathbb{R}^{d_1 \times d_2 \times d_4}$

• If  $A \in \mathbb{R}^{d_1 \times d_2 \times d_3}$  is multiplied with a matrix  $B \in \mathbb{R}^{d_3 \times d_4}$

$$\begin{bmatrix} [A_1]_{d_2 \times d_3} \\ [A_2]_{d_2 \times d_3} \\ \vdots \\ [A_{d_1}]_{d_2 \times d_3} \end{bmatrix} \cdot [B]_{d_3 \times d_4} = \begin{bmatrix} [A_1 B]_{d_2 \times d_4} \\ [A_2 B]_{d_2 \times d_4} \\ \vdots \\ [A_{d_1} B]_{d_2 \times d_4} \end{bmatrix}$$

• If instead  $A \in \mathbb{R}^{d_1 \times d_2 \times d_3}$  is multiplied with a vector  $v \in \mathbb{R}^{d_3}$ ,

$$\begin{bmatrix} [A_1] \\ [A_2] \\ \vdots \\ [A_{d_1}] \end{bmatrix} v = \begin{bmatrix} [A_1 v]_{d_2} \\ [A_2 v]_{d_2} \\ \vdots \\ [A_{d_1} v]_{d_2} \end{bmatrix} \in \mathbb{R}^{d_1 \times d_2} \quad (\text{or } \mathbb{R}^{d_1 \times d_2 \times 1})$$

### Reshape: "flatten" matrices.

How can we flatten arrays rowwise?

E.g.

$$\underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}}_{A_1} \rightarrow \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} ; \quad \underbrace{\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}}_{A_2} \rightarrow \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} ; \quad \underbrace{\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}}_{A_3} \rightarrow \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\underbrace{\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}}_{A_4} \rightarrow \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Notice that  $\beta = \{A_1, \dots, A_4\}$  form a basis in  $\mathbb{R}^{2 \times d}$ :

$$\text{for any } B = \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix}: B = b_{11} A_1 + b_{12} A_2 + b_{21} A_3 + b_{22} A_4$$

$\Rightarrow$  We obtain a representation of  $B$  in this new basis:  $[B]_{\beta} = \begin{bmatrix} b_{11} \\ b_{12} \\ b_{21} \\ b_{22} \end{bmatrix}$

$\Rightarrow$  The matrix representation of this transformation, in my basis  $\beta$ , is the identity matrix!  $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ .

What if we do instead (order changed)  $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \longrightarrow \begin{pmatrix} a \\ c \\ b \\ d \end{pmatrix}$

### One and multidimensional chain-rule

$$\text{If: } \mathbb{R}^n \xrightarrow{g} \mathbb{R}^k \xrightarrow{f} \mathbb{R}^m$$

$$\text{E.g. } \begin{pmatrix} x \\ y \end{pmatrix} \xrightarrow{g} \begin{pmatrix} x^2 + y^2 \\ 0 \\ 3x - y \end{pmatrix} \xrightarrow{f} \begin{pmatrix} (x^2 + y^2)^2 \\ 0 \\ \exp(3x - y) \end{pmatrix}$$

$\hookrightarrow$  what is  $f(x, y, z) = ?$

• Notation, instead of  $f', g'$ , we have  $\nabla f$  or  $\nabla g$ .

• If  $f = [f_1, f_2, \dots, f_m]$  we have

$$\nabla f = \begin{bmatrix} \nabla f_1^T \\ \nabla f_2^T \\ \vdots \\ \nabla f_m^T \end{bmatrix} \in \mathbb{R}^{m \times k}$$

$f_i(x_1, \dots, x_k)$

• Chain rule:  $(f \circ g)'(x) = f'(g(x)) \cdot g'(x) \longrightarrow \text{total size } (m, n)$

$\uparrow$   $f': \mathbb{R}^k \rightarrow \mathbb{R}^{m \times k}$

$\nwarrow$   $g': \mathbb{R}^n \rightarrow \mathbb{R}^{k \times n}$

Example:

$$\bullet g(x, y) = \begin{pmatrix} x^2 + y^2 \\ 0 \\ 3x - y \end{pmatrix} \Rightarrow g'(x, y) = \begin{bmatrix} \nabla g_1 \\ \nabla g_2 \\ \nabla g_3 \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x}(x^2 + y^2) & \frac{\partial}{\partial y}(x^2 + y^2) \\ 0 & 0 \\ \frac{\partial}{\partial x}(3x - y) & \frac{\partial}{\partial y}(3x - y) \end{bmatrix} = \begin{bmatrix} 2x & 2y \\ 0 & 0 \\ 3 & -1 \end{bmatrix} \in \mathbb{R}^{3 \times 2}$$

$$\bullet f(x, y, z) = \begin{pmatrix} x^2 \\ y \\ \exp(z) \end{pmatrix} \Rightarrow f'(x, y, z) = \begin{bmatrix} \nabla f_1 \\ \nabla f_2 \\ \nabla f_3 \end{bmatrix} = \begin{bmatrix} \frac{\partial}{\partial x} x^2 & \frac{\partial}{\partial y} x^2 & \frac{\partial}{\partial z} x^2 \\ \frac{\partial}{\partial x} y & \frac{\partial}{\partial y} y & \frac{\partial}{\partial z} y \\ \frac{\partial}{\partial x} e^z & \frac{\partial}{\partial y} e^z & \frac{\partial}{\partial z} e^z \end{bmatrix}$$

$$= \begin{bmatrix} 2x & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^z \end{bmatrix} \in \mathbb{R}^{3 \times 3}$$

Using the chain rule:

$$(f \circ g)'(x) = f'(g(x)) \cdot g'(x)$$

$$= f'(x^2 + y^2, 0, 3x - y) \cdot g'(x)$$

$$= \begin{bmatrix} 2(x^2 + y^2) & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & e^{3x - y} \end{bmatrix} \cdot \begin{bmatrix} 2x & 2y \\ 0 & 0 \\ 3 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 4x(x^2 + y^2) & 4y(x^2 + y^2) \\ 0 & 0 \\ 3e^{3x - y} & -e^{3x - y} \end{bmatrix}.$$

↳ there are algorithms that perform differentiation for us (approximately)

$$\text{e.g. BDF1: } y_{n+1} - y_n = h f(t_{n+1}, y_{n+1}) \quad [\text{Backwards Euler}]$$

## Lecture 4

**Subspace:** A set  $S \subset \mathbb{R}^n$  is called a subspace if:

- $u+v \in S \quad (\forall u, v \in S)$
- $\lambda u \in S \quad (\forall \lambda \in \mathbb{R}, \forall u \in S)$

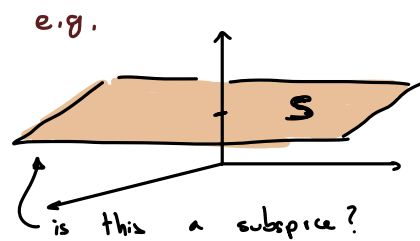
We can generate a subspace  $S$  with vectors  $b_1, \dots, b_n$  that are linearly independent

↳ We can't write any  $b_j$  as a linear combination of the others.

and for any  $u \in S$ :

(they span  $S$ )  $u = \sum_{i=1}^n \alpha_i b_i$  for some  $\alpha_i \in \mathbb{R}$ .  
↑ these are unique.

$\Rightarrow$   $\begin{cases} n \text{ is called the dimension of } S. \\ b_1, \dots, b_n \text{ is called a basis.} \end{cases}$



**Example:**

We know that a plane in  $\mathbb{R}^3$  can be describe by the equation:

$$ax + by + cz + d = 0$$

If we add the restriction that  $(0,0,0)$  satisfies this eq. (i.e. it's in the plane)

then  $d = 0$ . What are the basis vectors that generate this plane?

Notice that:

$$ax + by + cz = 0 \quad (\text{suppose } a \neq 0)$$

$$\Rightarrow x = -\frac{b}{a}y - \frac{c}{a}z$$

Now, we want to find  $x, y, z$  that solve this equation!

Observe:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -\frac{b}{a}y - \frac{c}{a}z \\ y \\ z \end{pmatrix} = y \begin{pmatrix} -\frac{b}{a} \\ 1 \\ 0 \end{pmatrix} + z \begin{pmatrix} -\frac{c}{a} \\ 0 \\ 1 \end{pmatrix} \quad \text{and } y, z \text{ are arbitrary.}$$

We can see that any vector in our plane (in other words, any vector satisfying our equation)

is a linear combination of  $b_1 = \begin{pmatrix} -b/a \\ 1 \\ 0 \end{pmatrix}$  and  $b_2 = \begin{pmatrix} -c/a \\ 0 \\ 1 \end{pmatrix}$



↳ we see the dimension of this subspace is 2.

We can generalize this and define a subspace:

$$S = \{x \in \mathbb{R}^n \mid Ax = 0\} \quad \text{for some } A \in \mathbb{R}^{n \times n}$$

↑ Is this a subspace?

$$\begin{aligned} \text{If } u, v \in S: \quad u+v \in S? & \text{ Yes! } & A(u+v) &= Au + Av = 0 \checkmark \\ & & Ax(u) &= xAu = 0 \checkmark \end{aligned}$$

This is called the null space of  $A$ .

• Notice that if  $A$  is a square matrix and inv.  $\Rightarrow$  only  $x=0 \in S$ , i.e.  $\dim(S) = 0$ .

• E.g.  $\underbrace{\begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}}_A \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \Leftrightarrow \left( \begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 3 & 2 & 1 & 0 \end{array} \right) \Leftrightarrow \left( \begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & -4 & -8 & 0 \end{array} \right) \Leftrightarrow \left( \begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 0 \end{array} \right)$

We can compute  $\text{null}(A)$  by performing Gaussian elimination (solve the equation!)

More geometric concepts:

Length of a vector:  $|v| = \sqrt{v \cdot v}$

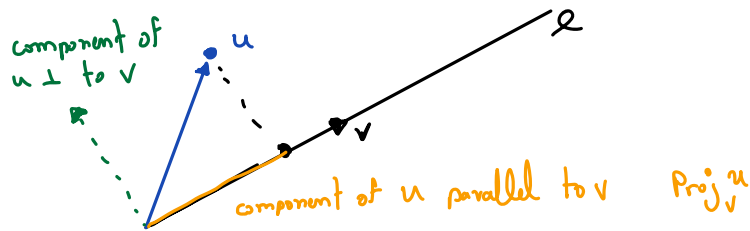
↑ dot product!

We say two vectors are orthogonal ( $u \perp v$ ) if  $u \cdot v = 0$



## Orthogonal decomposition:

Suppose we have a line spanned by  $v$



We can show:

$$u \cdot v = |u||v|\cos\theta$$

Proof: law of cosines



$$|u-v|^2 = |u|^2 + |v|^2 - 2|u||v|\cos\theta$$

$$\text{but } |u-v|^2 = (u-v) \cdot (u-v) = |u|^2 + |v|^2 - 2u \cdot v$$

$$\Rightarrow u \cdot v = |u||v|\cos\theta.$$

How can we find these components?

1. Since  $\text{Proj}_v u$  is parallel to  $v$ , there must be a constant s.t.  $\text{Proj}_v u = \alpha v$ .

2. The orthogonal component and  $v$  satisfy:

$$(\text{Proj}_v u)^\perp \cdot v = 0$$

3. We know  $u = \text{Proj}_v u + (\text{Proj}_v u)^\perp$ :

$$(u - \text{Proj}_v u) \cdot v = 0$$

4.  $\alpha v = \text{Proj}_v u$ :

$$(u - \alpha v) \cdot \alpha v = 0 \Rightarrow \alpha [u \cdot v - \alpha |v|^2] = 0$$

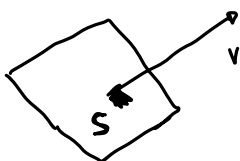
$$\Rightarrow u \cdot v = \alpha |v|^2$$

$$\Rightarrow \alpha = \frac{u \cdot v}{|v|^2}$$

$$\Rightarrow \text{Proj}_v u = \frac{u \cdot v}{|v|^2} v$$

orthogonal subspace: A subspace  $S \subset \mathbb{R}^k$  is orthogonal to  $v$  if for all

$$u \in S: v \perp u.$$



do we have to check  $v \cdot u = 0$  for every  $u$  in  $S$ ?

## Range and rank of matrices.

For large problems, certain matrix representations of linear maps can be unnecessarily expensive. For instance:

$$T: \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

$x \mapsto Tx$  is the projection of  $x$  onto  $v = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$

The matrix representation:

$$P = \begin{pmatrix} T(e_1) & T(e_2) & T(e_3) \end{pmatrix} = \begin{pmatrix} 1/2 & 0 & 1/2 \\ 0 & 0 & 0 \\ 1/2 & 0 & 1/2 \end{pmatrix}$$

However, we know that the rank of this matrix is 1, since the dimension span by its column space is 1!

→ **Rank:** The rank of a matrix  $A$  is the # of ind. row vectors.

Equivalently: # of independent column vector.

: The dimension of the subspace spanned by row vectors  
↑ or columns!

→ for  $A \in \mathbb{R}^{n \times k}$  we take an arbitrary vector  $x \in \mathbb{R}^k$   
and see the space  $Ax$  spans, its dimension will be  $= \text{rank}(A)$

**Remark:** If we know  $\text{rank}(A) \leq n$ , we can always find matrices

$$U \in \mathbb{R}^{n \times \text{rank}(A)}, \quad V \in \mathbb{R}^{k \times \text{rank}(A)}$$

s.t.  $A = UV^T$ , where  $\text{rank}(U) = \text{rank}(V) = \text{rank}(A)$ .

This means instead of using all  $n \cdot k$  entries of  $A$ , we only

need  $k \cdot \text{rank}(A) + n \cdot \text{rank}(A) = \text{rank}(A)^2$

• Going back to  $P = \begin{pmatrix} 1/2 & 0 & 1/2 \\ 0 & 0 & 0 \\ 1/2 & 0 & 1/2 \end{pmatrix}$ . We can write  $P$  as:

$$P = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 1/2 & 0 & 1/2 \end{pmatrix} = \begin{pmatrix} 1/2 & 0 & 1/2 \\ 0 & 0 & 0 \\ 1/2 & 0 & 1/2 \end{pmatrix}$$

## Projection matrix

Example: 3D cone. Suppose we have  $P: \mathbb{R}^n \rightarrow \mathbb{S} \subset \mathbb{R}^n$  that projects  $v \in \mathbb{R}^n$  onto a 3D subspace.

- Take a basis  $\{b_1, b_2, b_3\}$  of this subspace.
- Compute the projection of  $e_1, e_2, e_3$  onto this subspace.
- This can be given with 3 numbers:

$$Pe_i = a_i b_1 + a_2 b_2 + a_3 b_3$$

$\Rightarrow$  i.e. instead of storing  $n$ -components of  $Px$ ,  $x \in \mathbb{R}^n$ , we only need 3!

Then, the matrix representing  $P$ :

$$(Pe_1 \quad Pe_2 \quad \dots \quad Pe_n)$$

where each  $Pe_i$  belongs to  $\mathbb{S}$ , this matrix has size  $(3, n)$ .

Now, suppose  $\{b_1, b_2, b_3\}$  is orthonormal, i.e.,  $\|b_i\|^2 = 1$  and  $b_i \cdot b_j = \begin{cases} 1; i=j \\ 0; i \neq j \end{cases}$

It can be proved that

$$P = BB^T \quad \text{where} \quad B = \begin{pmatrix} b_1 & b_2 & b_3 \end{pmatrix} \in \mathbb{R}^{n \times 3}$$

$\hookrightarrow B^T$  maps  $\mathbb{R}^n$  to  $\mathbb{R}^3$  and then we reconstruct that back to  $\mathbb{R}^n$ .

How can we get  $\{b_1, b_2, b_3\}$  in practice?

•  $\hookrightarrow$  IF we already have 3 l.i. vectors, then we can obtain an orthonormal basis using a **Gram-Schmidt** procedure.

e.g. with 3 l.i. vectors  $v_1, v_2, v_3$ .

$$1. u_1 = v_1$$

$$2. u_2 = v_2 - \text{Proj}_{u_1}(v_2) \quad \left. \begin{array}{l} \text{Project } v_2 \text{ onto } u_1 \text{ and take} \\ \text{its difference} \end{array} \right\}$$

$$3. u_3 = v_3 - \text{Proj}_{u_1}(v_3) - \text{Proj}_{u_2}(v_3) \quad \left. \begin{array}{l} \text{Project } v_3 \text{ onto } \{u_1, u_2\} \\ \text{and take its difference.} \end{array} \right\}$$

• If we don't have l.i. vectors? Take vectors  $v_1, \dots, v_k$  s.t they span the subspace  $S$ , then create a matrix:

$$\begin{pmatrix} v_1 & v_2 & \dots & v_k \end{pmatrix}$$

Then, perform row-reduction (Gaussian elimination) to find the pivot columns.

### Stochastic matrices

• Properties: for  $A = (a_{ij})_{ij} \Rightarrow 1. a_{ij} \geq 0$

$$2. \sum_{i=1}^n a_{ij} = 1 \quad \text{for all } j$$

$$3. A \in \mathbb{R}^{n \times n} \quad (\text{square matrix})$$

Special case: double stochastic matrices:  $\sum_{j=1}^n a_{ij} = 1 \quad \text{for all } i.$

↳ Special case: Permutation matrices.

→ A permutation of  $n$  elements:  
 $(1 \ 2 \ 3 \ 4 \ 5 \ 6 \ \dots \ n)$

can be given by the image of all elements:

$$\begin{array}{ccccc} \text{e.g.} & (1 & 2 & 3 & 4 & 5) \\ & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow \\ & 2 & 3 & 5 & 1 & 4 \end{array}$$

$$\Rightarrow j\text{th element} \rightarrow \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$\uparrow$   $\uparrow$   $\uparrow$   $\uparrow$   $\uparrow$   $\uparrow$   
 $P\left(\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}\right)$   $P\left(\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}\right)$   $P\left(\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}\right)$   $P\left(\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}\right)$   $P\left(\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}\right)$   $P\left(\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}\right)$